A JEE Notebook

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Preface

The Philosophy

First, before jumping to any content, I shall discuss the philosophy behind these notes. These notes are made because I did not like pen-paper notes, these have version control, can be easily rewritten, and are more accessible(on phone/pc and if needed, on print).

Another thing to note is that these notes follow a specific philosophy, in that they try to be expository. They are meant to explain things. I have noticed that books often talk circularly, create redundancies, and etc.

To combat this, I have tried to clearly lay out everything here. Also, because of how books are made, generally in various volumes, content often goes repeated. These notes will try to remove that.

Lastly, I have often added some sort of additional non-JEE relevant content in many places, because 1) JEE is not the only thing I care for, 2) I am particularly interested in Mathematics/Physics so I study extra stuff, 3) They often help in exposition and better understanding. These are, however, mentioned in the text, and you can easily skip them.

If you find any typos, please contact me at `adyanshmishra@proton.me`, create a pull request, or dm me on discord at username Anonscholar.

Happy learning!

PART A

Algebra

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1 Sets

Motivation: There is little Mathematics that does not use the language of sets.

§1.1 Naive Set Theory

Sets are perhaps the most important structures in all of Mathematics. Although an indepth discussion of sets is beyond these notes, and their axiomation is complicated, we shall not let this affect us and discuss a less rigorous formalisatpion of sets. Namely, *Naive Set Theory*.

Definition 1.1.1

A set is any collection of elements.

It is denoted as $\{a, b, c\}$ where a, b, c are elements of the set. This representation of sets is called *Roster form*.

▶ These definitions are superfluous, we have not and will not define what we mean by collection or elements. That is what makes this theory *naive*.

Well, what do we mean by any collection? To give a few examples, the set of flowers would be {roses, lilies, sunflowers, ...}. The trailing "..." are used to show that the elements continue on *ad infinitum*. Hence, any collection of objects is a set.

Often, instead of manually writing out {roses, lilies, sunflowers, ...}, we will use a condition that each element in the set follows. Here, the condition is that all elements all flowers. We denoted this by saying that S is the set of all x such that x is a flower. We represent this in what is called *Set-Builder* notation. Thus,

$$S = \{x : P(x)\} \quad \text{or} \quad S = \{x \mid P(x)\}$$

Here, $P(x)$ represents the condition. We use the different symbols depending on the context, mostly for ensuring that the statements are clear.

Remark 1.1.2. Notice the equality symbol between S and $\{x : P(x)\}$. Generally, we denote sets as capital latin letters $A, B, C, \ldots$, and their elements by little latin letters $a, b, c, \ldots$.

Example 1.1.3

The sets of all Natural numbers is denoted by $\mathbb{N}$. Generally, it is either used to represent $\{0, 1, 2, \ldots\}$ or $\{1, 2, 3, \ldots\}$. It matters little. We will let $\mathbb{N} = \{1, 2, 3, \ldots\}$ and $\mathbb{N}_{\geqslant 0} = \{0, 1, 2, \ldots\}$ in this text

Example 1.1.4

The other common sets are similarly denoted by *blackboard bold* letters as well.

1. $\mathbb{Z}$ is the set of all integers.
2. $\mathbb{Q}$ is the set of all rationals.
3. $\mathbb{R}$ is the set of all reals.
4. $\mathbb{C}$ is set of all complex numbers.

▶ The symbol $\in$ in $x \in S$ is used to show that x is an element of S. It may be read as 'belongs to'.

Example 1.1.5

The cartesian plane is also a set. It defined as

$$\{(x, y) \mid x, y \in \mathbb{R}\}$$

§1.1.1 Extra Why Naive?

One interesting formulation possible in naive set-theory is the "Russel's Paradox". It goes as follows. Let S be a set $\{x \mid x \notin S\}$. Basically, S is the set of all elements not in S. This leads to a paradox because for x to be in S, it has to follow the condition that it is not in S! And any element not in S must be in S.

There is no particular solution to this using naive set-theory. This is one of the reasons why this discussion of sets is so naive.

§1.1.2 Types of Sets

Although this division of sets is superfluous at best, we will still categorise them for a reference.

▶ The set $\{\varnothing\}$,is not a null set! In-fact it is the set that contains the empty or null set.

- **Finite Set** : A set having a finite number of elements is called a finite set.
- **Infinite Set** : A set which has infinite elements is called an infinite set.
- **Null Set** : A set which has no elements is called a null set. It is represented as $\{\}$ or $\varnothing$.
- **Singleton Set** : A set containing a single element is called a singleton set.

§1.2 Equality and Subsets

§1.2.1 Sets of Sets

Remember that we referred to sets as a collection of *any* objects? That might lead to the question that can we also make sets of sets? In-fact, that is exactly right, we can! A set of sets maybe represented as $\{\{a, b, c\}, \{d, e, f\}, \{g, h, i\}\}$. All of the elements of this set are themselves sets. One thing to note here is that we cannot form a set of all sets, and we will see why in a moment.

§1.2.2 Subsets

▶ The empty set $\varnothing$ is a subset of all sets.

Consider the sets $\mathbb{N}$ and $\mathbb{Z}$. You might see that all the elements of $\mathbb{N}$ are also present in $\mathbb{Z}$. If we were to describe it, we could say that $\mathbb{N}$ is contained in $\mathbb{Z}$. Such a relation between sets is expressed by saying $\mathbb{N}$ is a subset of $\mathbb{Z}$.

Definition 1.2.1

A set A is a subset of B if all elements of A are also elements of B.
This is represented as

$$A \subseteq B$$

Example 1.2.2

$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$.

Example 1.2.3

Let $A = \{x \mid x \text{ is prime and } x \text{ is odd}\}$, and $B = \{x \mid x \text{ is prime}\}$, then $A \subseteq B$.

One thing to note here is that for A to be a subset of B, it does not need to have less elements than B. Infact, A is a subset of A itself because all elements of A are in A.

To show that each element of A is contained in B but all elements of B are not contained in A, we say that A is a proper subset of B.

Definition 1.2.4

A is a proper subset of B if all elements of A are in B but all elements of B are not in A.
We represent this as

$$A \subsetneq B$$

Note that A is not a proper subset of A.

§1.2.3 Superset

If A is a subset of B, then B is a superset of A. It is represented as

$$B \supseteq A$$

This is another way of saying the same thing. The definition of proper superset is also the same.

§1.2.4 Equality

We may say that two sets are equal if they have the same elements. However, there is one thing to consider. When discussing about sets, we do not care how many times an element appears. The sets $\{1, 1, 1, 1, \dots\}$ is the same as $\{1\}$. Therefore, we always ignore the repeating terms. Thus, two sets are equal if they have the same number of elements, **regardless of the frequency they appear in.** The same thing applies for subsets and supersets, **the frequency of elements does not matter.**

We can phrase the equality of sets in another manner, in terms of subsets.

Definition 1.2.5 (Equality in sets)

Two sets A and B are equal if and only if $A \subseteq B$ and $B \subseteq A$.
Therefore,

$$A = B \iff A \subseteq B \text{ and } B \subseteq A$$

Refer to Chapter A for the symbol $\iff$.

§1.2.5 Power-set and Cardinality

The powerset of A is represented as $P(A)$ or 2^A. It is the set of all subsets of A. Note that it is the set of all *subsets* and not proper subsets, therefore, A is an element of $P(A)$. Formally,

▶ Because of the definition of powerset, $A \subseteq B \iff P(A) \subseteq P(B)$.

Definition 1.2.6 (Power Set)

$P(A)$ is called the powerset of A, and

$$P(A) = \{X \mid X \subseteq A\}$$

Cardinality

Vaguely, cardinality is the size of a set. It is the number of elements of a set. The cardinality of a set is represented as $n(A)$.

Definition 1.2.7

The cardinality of a set A, $n(A)$ is

$$n(A) = \text{ Number of elements in } A$$

§1.2.6 Universal Set

The Universal Set is the set consisting of all elements in consideration. All sets which we consider are its subsets. Note that the universal set is arbitrary. We can choose any set as the universal set. Generally, using a universal set allows us to draw Venn diagrams. I won't discuss venn diagrams here though. Even though they're intuitive, there is little use of describing them.[1]

§1.3 Operations on Sets

§1.3.1 Union of Sets

▶ 'Or' in mathematics refer to something along the lines of one, the other, or both. This is in contrast with English, in which it refers to one or the other. The case of both is also included by 'or' in mathematics.

Definition 1.3.1 (Union of Sets)

The union of two sets, A, B is the set of all elements in A or B.

$$A \cup B = \{a \mid a \in A \text{ or } a \in B\}$$

Example 1.3.2

If $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$. Then $A \cup B = \{1, 2, 3, 4, 5, 6\}$.

The union of two sets can be understood as something like 'addition'. $A \cup B$ can be naively thought to 'add' the elements in both sets. Since we do not care about the multiplicity (frequency) of the elements, repeating elements are just counted once. This is another way of saying 'or' as we had previously. There are some properties of union, none too difficult to grasp, to consider.

[1] If you wish to study them, look them up online, there is a lot of sources that will adequately satisfy you.

Proposition 1.3.3 — Yeah

§1.3.2 Cartesian Product

The cartesian product of two sets, A and B is defined as,

$$A \times B = \{(x, y) \mid x \in A, y \in B\}$$

The use of cartesian product is mainly for ordered pairs. The nth cartesian product of a set is represented as,

$$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ times}}$$

2 Elementary Prereqs

§2.1 Intervals

Definition 2.1.1

For $a, b, x \in \mathbb{R}$ and $a < b$,

1. The subset $\{x \mid a < x < b\}$ of $\mathbb{R}$ is called the open interval of a, b and is represented as (a, b) or $]a, b[$.
2. The subset $\{x \mid a \leqslant x \leqslant b\}$ of $\mathbb{R}$ is called the closed interval of a, b and is represented as $[a, b]$.

▶ The notation $]a, b[$ is French, and is not used in India, USA, etc.

Add figures for all

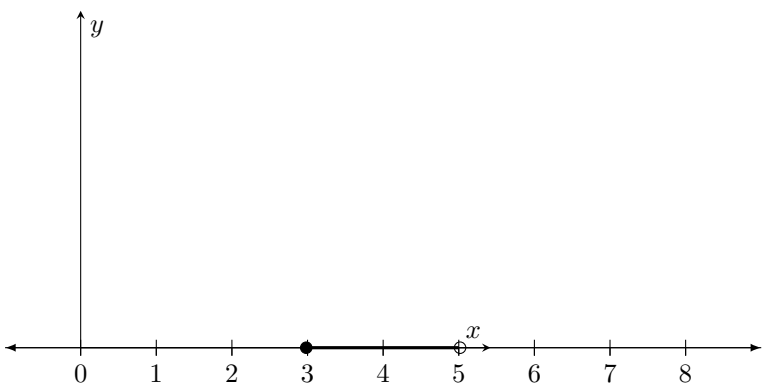

Figure 2.1: Closed-Open Interval

We can make all kind of closed-open ($a \leqslant x < b$) or open-closed ($a < x \leqslant b$) intervals, but there is no point in giving them a name.

§2.2 Inequalities

Real numbers and all its subsets have a common property. They're ordered under the $<, >, =$ relations. Thus, for any two real numbers, we can always compare them and have the trichotomy :

Claim 2.2.1 — For $a, b \in \mathbb{R}$ we will always have,

1. $a < b$,

2. $a = b$, or
3. $a > b$.

This why inequalities are so useful for reals! We can always form inequalities for equations. Now, let us propose the following properties of inequalities.

▶ An inequality or in-equation of x describes a property of x such that the answer to it is generally an interval or a set of possible values which might satisfy it. In contrast to equations, it rarely gives discrete answers, and instead of equality, we have the ordering $(<, >, \geqslant, \leqslant)$ relations.

Proposition 2.2.2 — For $a, b \in \mathbb{R}$,

1. $a < b \iff a \pm k < b \pm k$
2. $a < b, c < d \iff a + c < b + d$
3. $a < b, c < d \iff a - d < b - c$
4. $a < b, k > 0 \implies ka < kb$
5. $a < b, k < 0 \implies ka > kb$
6. $a < b, b < c \implies a < c$
7. $0 < a < b, r > 0 \implies a^r < b^r$
8. $0 < a < b, r < 0 \implies a^r > b^r$
9. $a > 0 \iff a + \dfrac{1}{a} \geqslant 2$
10. $a < 0 \iff a + \dfrac{1}{a} \leqslant 2$

§2.2.1 Inequalities concerning inverses

If $x \in (a, b)$, then,

Case I $ab > 0$

If a and b have the same sign, then,

$$\frac{1}{x} \in \left(\frac{1}{b}, \frac{1}{a}\right)$$

Case II $ab < 0$

If a and b have different signs then,

$$\frac{1}{x} \in \left(-\infty, \frac{1}{b}\right) \cup \left(\frac{1}{a}, \infty\right)$$

§2.2.2 Inequalities concerning squares

One thing of note to consider is how do the inequalities of x from x^2 follow? Consider $a \leqslant x \leqslant b$. You might expect that for $a \leqslant x \leqslant b$, $a^2 \leqslant x \leqslant b^2$. But that is simply incorrect! What we instead get is that x^2 lies between 0 and $\max(a^2, b^2)$. Here, max represents that x^2 lies between 0 and whichever of the a^2, b^2 is greater. In-fact, if $a \leqslant x \leqslant b \iff a^2 \leqslant x^2 \leqslant b^2$, then a, b and hence x must be $\geqslant 0$.

Proposition 2.2.3 — For $x, a \in \mathbb{R}$,

- If $x^2 \leqslant a^2$, then $x \in [-a, a]$.
- And, if $x^2 \geqslant a^2$, then $x \in (-\infty, -a] \cup [a, \infty]$.

Example 2.2.4

For $-5 \leqslant x \leqslant 4$, we have $0 \leqslant x^2 \leqslant \max(-5^2, 4^2)$. Therefore, $0 \leqslant x^2 \leqslant 25$.

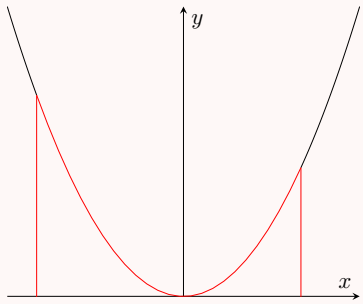

Figure 2.2: x^2 for $-5 \leqslant x \leqslant 4$

§2.3 Sign Scheme Method

A way of determining inequalities of the form $F(x) = (x - a_1)^{k_1}(x - a_2)^{k_2} \dots (x - a_n)^{k_n}$ where $F(x) \leqslant 0$, or $F(x) \geqslant 0$ is the *Sign Scheme Method*. Arrange $a_1, a_2, \dots, a_n$ such that $a_1 \leqslant a_2 \leqslant \dots \leqslant a_{n-1} \leqslant a_n$. Now follow the following steps:

1. Mark $a_1, a_2, \dots, a_n$ on the number line, and put the plus sign on the right of the largest of these, a_n. Since when $x \geqslant a_n$, $x \geqslant a_1, a - 2, \dots$. So $F(x) \geqslant 0$.
2. Put the plus sign on the interval to left of a_n if k_n is even, and the minus sign if it is odd. Similarly, when passing to the left of a_i, plus sign if k_i is even, minus otherwise.
3. The union of all intervals with the plus sign is the solution for $F(x) \geqslant 0$, and the union of the ones with minus sign, for $F(x) \leqslant 0$.

Example 2.3.1

Consider $\dfrac{2y-1}{y} \leqslant 0$. First, we change it into the already known form,

$$\frac{2y-1}{y} \leqslant 0 \tag{2.1}$$

$$2\frac{(y-1/2)}{y} \leqslant 0 \tag{2.2}$$

$$\frac{y-1/2}{y} \leqslant 0 \tag{2.3}$$

$$(y-1/2)(y-0)^{-1} \leqslant 0 \tag{2.4}$$

Plotting these we get,

Figure 2.3: Sign Scheme Method in action

Since We are concerned with $\dfrac{2y-1}{y} \leqslant 0$ and can't have $y = 0$, $y \in \left(0, \dfrac{1}{2}\right]$.

3 Complex Numbers

We will begin our study of complex numbers in a little bit of a different manner. Instead of using algebraic form as the base, we will first consider the set $\mathbb{R}^2$,

$$\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$$

Clearly, $(x_1, y_1) = (x_1, y_1) \iff x_1 = x_2, y_1 = y_2$ by the definition of ordered pairs.

Let us represent the elements as $z = (x, y)$. Now, let us define the operations of addition and multiplication on this set as,

$$z_1 + z_2 = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

And multiplication is defined as,

$$z_1 \cdot z_2 = (x_1, y_1) \cdot (x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1)$$

Although the definition of multiplication may seem complicated now, we will be able to gain intuition for it soon.

▶ The letter z is used to represent a complex number.

Definition 3.0.1 (Complex Numbers)

An ordered pair, (x, y) such that,

$$z = (x, y) \in \mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$$

with the operations of addition and multiplication as defined above is called a *complex number*.

The field of complex numbers is the earlier set defined with the earlier operations, $(\mathbb{R}^2, +, \cdot)$.

Remark 3.0.2. Note that,

- $z_1 = (x_1, 0)$, $z_2 = (x_2, 0)$. Then, $z_1z_2 = (x_1x_2, 0)$.
- $z_1 = (0, y_1)$, $z_2 = (0, y_2)$. Then, $z_1z_2 = (-y_1y_2, 0)$.

The usual properties regarding addition and multiplication, commutativity, associativity, and existence of identity and inverses are true, and can be shown through simple proofs by properties of reals.

§3.1 Inverses

The additive inverse of $z = (x, y)$ is defined $-z = (-x, -y)$.

The multiplicative inverse of z, z^{-1}, such that $zz^{-1} = 1$ is, $(x, y) \cdot (a, b) = (1, 0)$. We get that,

$$\begin{cases} ax - yb = 1 \\ ay + bx = 0 \end{cases}$$

And we simply have,

$$\frac{1}{z} = z^{-1} = \left(\frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2} \right)$$

Two complex number z_1, z_2 also *uniquely* determine a quotient,

$$\frac{z_1}{z_2} = z_1 z_2^{-1} = \left(\frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2}, \frac{-x_1 y_2 + y_1 x_2}{x_2^2 + y_2^2} \right)$$

The set $\mathbb{R}^2$ with the operations addition and multiplication forms the field of complex numbers, $\mathbb{C}$.

Powers are defined similar to reals,

$$z^n = \underbrace{z \cdot z \cdot \cdots \cdot z}_{n \text{ times}}$$

for all integers $n > 0$. For $n < 0$, $z^n = (z^{-1})^{-n}$.

▶ We shall use $\mathbb{C}^*$ to show $\mathbb{C} \setminus \{0\}$.

Proposition 3.1.1 — For $z_1, z_2, z \in \mathbb{C}^*$, $a, b \in \mathbb{Z}$,

- $z^m \cdot z^n = z^{m+n}$.
- $\dfrac{z^m}{z^n} = z^{m-n}$.
- $(z^m)^n = z^{mn}$.
- $(z_1 z_2)^n = z_1^n \cdot z_2^n$.

§3.2 Algebraic Form of Complex Numbers

To deal with complex numbers, we introduce their algebraic form. Consider a function f,

$$f \colon \mathbb{R} \to \mathbb{R} \times \{0\}, f(x) = (x, 0)$$

Clearly such a f is a bijective. Note that by using our earlier definitions of addition and multiplication, $(x, 0)$ behaves exactly like x.

Therefore, we may write that $\forall (x, 0) \in \mathbb{R} \times \{0\}, \exists x \in \mathbb{R}$ such that we can write $(x, 0)$ as x.

Definition 3.2.1

We define $i \in \mathbb{C}$ as,

$$i = (0, 1) \in \mathbb{C}$$

By the definition of multiplication and using f as defined earlier, we may get that

$$\begin{aligned} z &= (x, y) = (x, 0) + (0, y) \\ &= (x, 0) + (y, 0) \cdot (0, 1) \\ &= x + yi = x + iy \end{aligned}$$

This gives us a fascinating result! We can represent each of the complex numbers in a simple algebraic form.

Lemma 3.2.2

Every complex number, (x, y) can be represented as

$$z = x + yi \mid x, y \in \mathbb{R}$$

where x is the real part of z and is denoted $\mathfrak{R}(z)$. y is said to be the imaginary part of z and is denoted $\mathfrak{I}(z)$.

$i^2 = -1$ by definition, since, $i \cdot i = (0, 1) \cdot (0, 1) = (-1, 0) = -1$.

This representation of complex numbers is called the algebraic representation, where we can say that $\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}, i^2 = -1\}$.

Complex numbers of the form $0 + iy$ are purely imaginary and the ones of the form $x + 0i$ are purely real.

Using an ordered pair first approach allows us to discern something very interesting, the difference between i and $-i$. Clearly, both are roots of $x^2 + 1$. However, it is rather difficult to differentiate them. Let's say $i = \sqrt{-1}$ and thus, $-i = -\sqrt{-1}$. What exactly is the difference between them?

The answer is that it is not so obvious. We can easily differentiate between $\sqrt{a}$ and $-\sqrt{a}$ for any real a the former is positive and the latter negative, because $\sqrt{\bullet} : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$, the principal square root function, is defined to return a positive value.

Unfortunately, no such thing is generally true for $\mathbb{C}$. The complex numbers aren't "ordered" like the reals. You can't establish $z_1 \geqslant z_2$ in general, except for the special case when they are both purely real. So how exactly are they different?

If we don't define i to $(0, 1)$ and instead $\sqrt{-1}$, we can't establish a difference. After all, why would i lie on the positive y axis and not the negative one? In reals, we could this would be true if $i \geqslant 0$. I mean, isn't that how we plot stuff on cartesian plane anyway?

But here, the square root definition restricts us a lot. We can't know by it if i is $(0, 1)$ or $(0, -1)$. Of course, we can define our axes such that it automatically lies on the positive y axis, but I feel that the ordered pair definition is much better, since there we know the difference i and $-i$. The former is $(0, 1)$ and the later $(0, -1)$.

And clearly, both are still solutions of $x^2 + 1 = 0$! Now of course, we can define the principal square root, $\sqrt{-1}$ to be i but in general, it is better to avoid it all together because there is little consensus on which of the two complex roots of such a polynomial should be the principal root.

§3.2.1 Operations in algebraic form

The addition of two complex numbers in algebraic form is simply,

$$(x + iy) + (a + ib) = (x + a) + i(y + b)$$

Thus,

$$\mathfrak{R}(z_1 + z_2) = \mathfrak{R}(z_1) + \mathfrak{R}(z_2)$$

Multiplication while not much more intuitive yet, is easy to calculate using $i^2 = -1$.

$$(x + iy)(a + ib) = (xa - yb) + i(xb + ay)$$

Which gives us,

$$\mathfrak{R}(z_1 z_2) = \mathfrak{R}(z_1)\mathfrak{R}(z_2) - \mathfrak{I}(z_1)\mathfrak{I}(z_2)$$
$$\mathfrak{I}(z_1 z_2) = \mathfrak{R}(z_1)\mathfrak{I}(z_2) - \mathfrak{R}(z_2)\mathfrak{I}(z_1)$$

§3.2.2 Powers of i

Using some simple algebraic manipulation,

$$\begin{aligned} i = i, \quad i^2 = -1, \quad i^3 = i^2 \cdot i = -i, \quad i^4 = (i^2)^2 = 1; \\ i^5 = i^4 \cdot i = i, \quad i^6 = i^4 \cdot i^2 = -1, \quad \ldots \end{aligned}$$

Thus, for $n \in \mathbb{Z}$,

$$i^{4n} = 1, \quad i^{4n+1} = i, \quad i^{4n+2} = -1, \quad i^{4n+3} = -i.$$

Also, we have,

$$i^{4n} + i^{4n+1} + i^{4n+2} + i^{4n+3} = 0$$

§3.3 Complex Conjugate

Definition 3.3.1

The complex conjugate, $\overline{z}$ of $z = x + iy$ is,

$$\overline{z} = x - iy$$

We can immediately get some properties of complex numbers, mainly,

Proposition 3.3.2 — For $z \in \mathbb{C}$, and $\overline{z}$ the complex conjugate of z, we have,

1. $z = \overline{z} \iff z \in \mathbb{R}$.
2. $z = -\overline{z} \iff z \in \mathbb{C} \setminus \mathbb{R}$
3. $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
4. $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$
5. $\overline{z^{-1}} = \overline{z}^{-1} \mid z \neq 0$
6. $\Re(z) = \dfrac{z + \overline{z}}{2}$, $\Im(z) = \dfrac{z - \overline{z}}{2i}$

PART B

Calculus

Part B: Contents

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4 Functions and Relations

§4.1 Relation

A relation between two sets A, B is the subset of their *cartesian product* $A \times B$. Since, a cartesian product between A and B is also a set of the ordered pairs, so is a relation R,

§1.3.2 Cartesian Product

$$R \subseteq \{(a, b) \mid a \in A, b \in B\}$$

There need not be any specific relation between the two elements. But often, it is useful to talk about relations which employ a specific relationship between the two elements. Hence, we can think of a relation as somehow connecting the two elements.

A relation on a set A is simply the subset of $A \times A$.

Example 4.1.1

Let $A = \{1, 2, 8\}$, $B = \{3, 5, 6\}$. Let R be the relation,

$$R = \{(a, b) \mid a > b, a \in A, b \in B\}$$

Then R contains the ordered pairs $(8, 3)$, $(8, 5)$ and $(8, 6)$

We show that $(a, b) \in R$ by saying aRb.

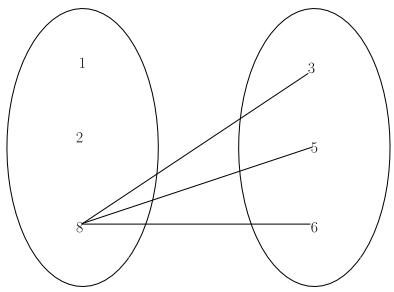


Figure 4.1: Pictorial representation of example 4.1.1

§4.1.1 Domain

The domain of a relation $R \subseteq A \times B$ is the set of all first elements of the ordered pairs of the relation, that are in set A.

$$D_R = \{a \mid (a, b) \in R\}$$

§4.1.2 Range and Co-domain

Range

Range of R is the set of all second elements of the ordered pairs in the relation, that are in set B.

$$Range_R = \{b \mid (a, b) \in R\}$$

> You may note that $Range_R \subseteq Co-domain_R$

Co-domain

Co-domain of $R \subseteq A \times B$ is the set B itself.

§4.2 Types Of Relations

Although it might seem like discussing the type of relations would be as trivial as sets, there are actually a couple really important type of relations. I'll mention the ones we really need to know about.

§4.2.1 Trivial Types

Mainly, these are trivial in the way that they're of very low importance to us. The ones that are useful later, have been especially marked.

Empty Relation

The relation R is an empty relation if it does not contain any elements. $R = \varnothing \subsetneq A \times B$.

Universal Relation

Yes, it is the relation that contains all the elements. A universal relation R on A is $R = A \times A$.

Inverse Relation

Let R be the relation from A to B. Then R^{-1} is the relation from B to A such that,

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}$$

Therefore, we may have $(a, b) \in R \iff (b, a) \in R^{-1}$.

Identity Relation

The identity relation I_A on A is defined as

$$I_A = \{(a, a) \mid a \in A\}$$

§4.2.2 Equivalence Relation

Reflexive relation

> All identity relations are reflexive relations but all reflexive relations are not identity relations. See example 4.2.1.

A relation R is a reflexive relation on A, if for every $a \in A$, $(a, a) \in R$.

Example 4.2.1

The relation $I_A = \{(1, 1), (2, 2), (3, 3)\}$ is an identity relation on the set $A = \{1, 2, 3\}$. However, the relation $R = \{(1, 1), (2, 3)(3, 2), (2, 2), (3, 3)\}$ is not an identity relation on A, but rather a reflexive relation. But $I_A \subsetneq R$.

Here, R is a reflexive relation but not an identity relation.

Symmetric Relation

A relation R on A is symmetric if $(a, b) \in R \iff (b, a) \in R$.

Example 4.2.2

Let R be the relation on $\mathbb{R}$ such that

$$R = \{(x, y) \mid x + y = 5\}$$

This relation is symmetric, because suppose any two reals x and $y \in R$. Then, $x + y = 5$. But since addition is commutative, $x + y = y + x = 5$. Thus, $(x, y) \in R \iff (y, x) \in R$.

Transitive Relation

A relation R on A is transitive if $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$.

Example 4.2.3

The relation $<$ is a transitive relation on $\mathbb{R}$. For instance, if $a < b$, $b < c$, then $a < c$.

Equivalence Relation

An equivalence relation on A is a relation that is transitive, reflexive, and symmetric.

Example 4.2.4

The relation '$=$' is an equivalence relation on $\mathbb{R}$. It is symmetric as $a = b \iff b = a$. It is reflexive as $a = a$. It is transitive as $a = b, b = c \implies a = c$.

▶ If you haven't already noticed, I have deliberately chosen examples that are non-conventional. I have done it in order to showcase that relations can be anything!

§4.3 Functions

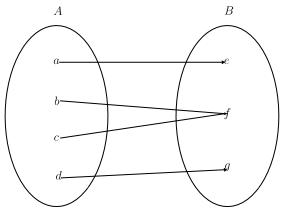
Functions are a special type of relation. They're, for most cases, much more useful than relations for us.

Definition 4.3.1

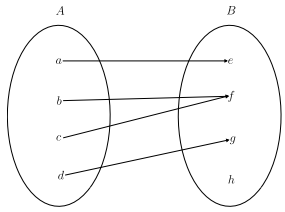
A function from A to B is a set of ordered pairs,

$$f \subseteq \{(a, b) \mid a \in A, b \in B \wedge (x, y) \in f, (x, z) \in f \iff y = z\}$$

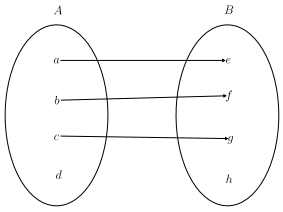
A little more informally, functions are a mapping from set A to B. We represent this by saying $f\colon A \to B$. When we map $a \in A$ to $b \in B$, we say that b is the 'image' of a under f and b is called the 'preimage' of a under f. We may represent that f maps a to b as $a \mapsto b$. Alternatively, and more commonly, $f(a) = b$.



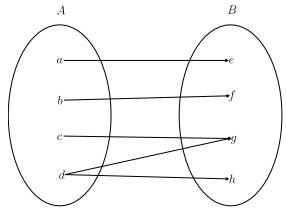
(a) A function from A to B.



(b) Also a function, note that h doesn't have a pre-image.



(c) Not a function, d has no image.



(d) Also not a function because d has two images.

Figure 4.2: Some examples of functions and not functions.

§4.3.1 Domain and Range

Domain

Although it may not be obvious, from definition 4.3.1 you may realise that every element of A has an image, i.e. every element in A is included in the ordered pairs of a function f. Thus, the domain of a function $f : A \to B$ is the whole set A.

Range and Co-domain

For a function $f : A \to B$, the co-domain is the set B. The Range of f is,

$$R_f = \{(x, f(x)) \mid x \in A\}$$

Example 4.3.2

For $f\colon \mathbb{R} \setminus \{0\} \to \mathbb{R}$ such that $f(x) = \dfrac{1}{x}$. The Range is $\mathbb{R} \setminus \{0\}$, as is the domain. The Co-domain is $\mathbb{R}$.

§4.3.2 Extra Questions of Domain and Range

There are many questions regarding the domain and range of a function, given an analytic formula of the same. These questions are weird in general, a function is categorised by not only its analytic formula[1], but also by its mapping.

For example, given $f(x) = e^x$, we may create infinite such functions f, the perhaps most natural one, $f : \mathbb{R} \to \mathbb{R}$, the not so obvious one $f : \mathbb{R} \to (0, \infty)$ and another not intuitive one, $f : \{1\} \to \{e\}$.

All of these are different from each other[2].

How could then we, given such a formula conjure up a domain and range? Well, by using some rather ill conventions. Generally, we find the domain that is the largest subset of $\mathbb{R}$ and the range that is defined by it. However, for functions of the form $h(x) = f(x)^{g(x)}$, we always assume $f(x) \geqslant 0$.

This leads to a nice domain but is rigorously inconsistent. I won't mention how to figure out these domain and ranges, it is not explicitly needed to be mentioned.

[1] Which it may or may not have.
[2] In fact, the second one is rather special, since it is bijective, something covered a little later.

§4.3.3 Informal Definition

A function can also be thought of a machine that takes in an independent variable and churns out a dependent variable. This is a very vague abuse of already known ideas to somehow understand the abstractness of the idea of a function.

$$\overset{\text{Independent}}{f(x)} = \underset{\text{Dependent}}{y}$$

Figure 4.3: Dependent and Independent variables

§4.3.4 Implicit and Explicit

Explicit Function

When we can clearly write out the dependent variable in terms of the independent variable, the function is called Implicit.

Example 4.3.3

$f(x) = y = x^3$ is an explicit function since it can be written with only the help of x.

Implicit Function

These are the functions in which the dependent variable cannot be written only in the terms of the independent variable.

Example 4.3.4

$x^3y^2 + x - y + xy = 0$ is an Implicit function. It may be represented as $R(x, y) = 0$.

§4.3.5 Digression
Extra

The functions we have considered till now are only real valued, and we have confined ourselves to a single-variable being independent. Really, the title of this part is wrong and should be more accurately Single-Variable Calculus.

The notions of implicit and explicit are quite annoying too, which are better dealt with in Multi-Variable Calculus. We will not release ourselves of this confinement for the rest of the part as it is irrelevant for JEE. But beware, you will know only a half of only Calculus even after JEE, let alone anything in Mathematics.

§4.3.6 Bounded Functions

A function is bounded if it is bounded from both above and below. A real-valued function, $f : A \to B$ is bounded from above if for each $a \in A$, $f(a) \leqslant M$. It is bounded from below if for each $a \in A$, $f(a) \geqslant N$.

Thus, a bounded function f lies in the interval $[M, N]$. It may also lie on any interval of the same kind, i.e (M, N), $[M, N)$, or $(M, N]$.

Example 4.3.5 (Bounded Functions)

Some examples of bounded functions are:

1. $\sin(x), \cos(x) \in [-1, 1]$
2. $\dfrac{1}{x^2} \in (0, \infty)$

Example 4.3.6 (Unbounded functions)

Some examples of unbounded functions are:

1. $\csc(x), \sec(x) \in (-\infty, -1] \cup [1, \infty)$
2. $\tan(x), \cot(x) \in (-\infty, \infty)$

§4.4 The "..." function

Extra

From this point we make many mentions of the "sine" function, the "cosine" function, and so on. However, there is no "sine" function. People generally call the function f, such that $f(x) = \sin(x)$ as the sine function. What they really mean is the function that maps x to $\sin(x)$. However, because of lack of notation and convenience, we refer to it as the "sine" function which is just an abuse of notation. "$\sin(x)$" is not a function. It is just a number.

§4.5 Graphing the Function

Consider any function f that maps from x to y, where any a is literally anything. It could be x^3, $\sin(x)$, or e^x. Basically, $y = f(x)$. If we map x and y as co-ordinates on the cartesian plane, we get the graph of the function, f such that $f(x) = y$.

Formally, a graph of a function f, Γ_f, is

$$\Gamma_f = \{(x, f(x)) \mid x \in D_f\}$$

where D_f is the domain of f.

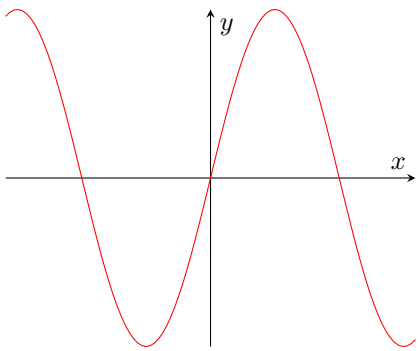
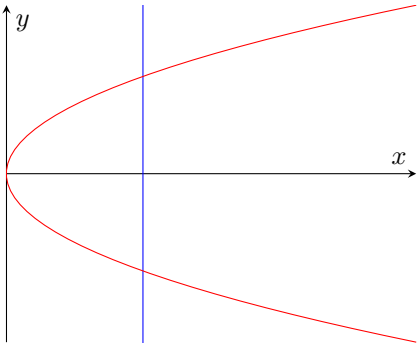


Figure 4.4: Graph of the function "$\sin(x)$"

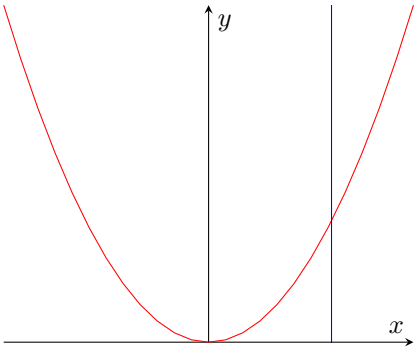
§4.5.1 Vertical Line Test

When we map f onto the cartesian plane, because of the very definition of a function, there are no two ordered pairs (x, a) and (x, b) since x cannot have two images. Because of this, if we draw a vertical line at any x for the function f, it must pass only once (since there is only

y for one x). We can use this test to determine through the graph of any relation, whether it is a function or not. This is called the vertical line test.



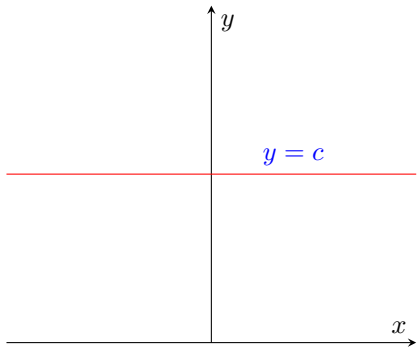
(a) Not a function



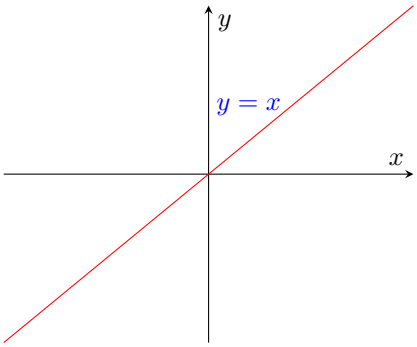
(b) A function

Figure 4.5: Vertical Line test for functions

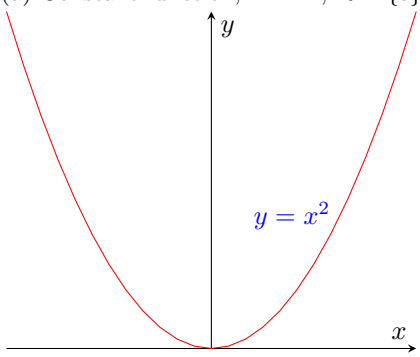
Here are the graphs of some elementary algebraic functions. Their domain and range are listed in the caption, where D is the domain, and R the range.



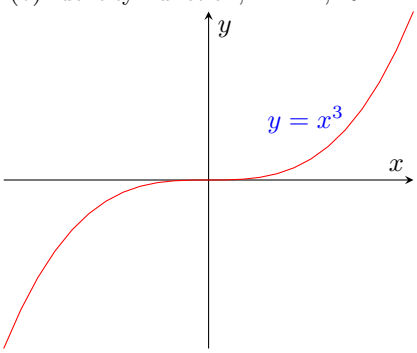
(a) Constant function, $D = \mathbb{R}$, $R = \{c\}$



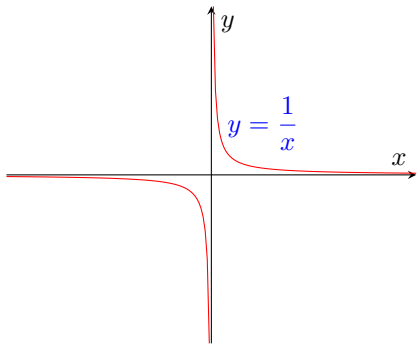
(b) Identity Function, $D = \mathbb{R}$, $R = \mathbb{R}$



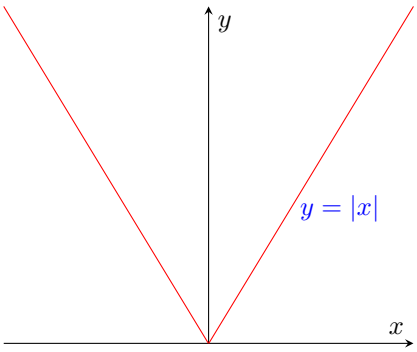
(c) Square Function, $D = \mathbb{R}$, $R = \mathbb{R}^+ \cup \{0\}$



(d) Cubic Function, $D = \mathbb{R}$, $R = \mathbb{R}$



(e) Reciprocal Function, $D = \mathbb{R} \setminus \{0\}$, $R = \mathbb{R} \setminus \{0\}$



(f) Modulus Function, $D = \mathbb{R}$, $R = \mathbb{R}^+ \cup \{0\}$

Figure 4.6: Some Elementary Functions

§4.6 Identical Functions

Two functions, f, g are said to be identical if and only if,

- Domain of both functions are equal, $D_f = D_g$.
- Range of both functions are equal, $R_f = R_g$.
- $\forall x \in D_f$ or D_g, $f(x) = g(x)$.

Example 4.6.1

$f(x) = x$ and $g(x) = \dfrac{1}{1/x}$ are identical for $x \in \mathbb{R} \setminus \{0\}$.

§4.7 Piecewise functions

A piecewise functions is a function f whose domain is partitioned into several intervals on which it maybe defined differently. We represent it as,

$$f = \begin{cases} F & \phi(x) \\ G & \varphi(x) \\ \vdots & \vdots \\ Z & \pi(x) \end{cases}$$

Where each of the $\phi(x)$, $\varphi(x)$, $\ldots$, $\pi(x)$ are some intervals of $x \in D_f$.

Example 4.7.1

The modulus function, $y = |x|$ is a piecewise function:

$$\begin{cases} x & x > 0 \\ 0 & x = 0 \\ -x & x < 0 \end{cases}$$

§4.8 Algebra of Functions

Let $f : A \to \mathbb{R}$, $g : A \to \mathbb{R}$ be two functions where $A \subsetneq \mathbb{R}$. Then, we have,

1. $f + g : A \to \mathbb{R}$, $(f + g)(x) = f(x) + g(x)$, for $x \in A$.
2. $f - g : A \to \mathbb{R}$, $(f - g)(x) = f(x) - g(x)$, for $x \in A$.
3. $f \cdot g : A \to \mathbb{R}$, $(f \cdot g)(x) = f(x) \cdot g(x)$, for $x \in A$.
4. $\dfrac{f}{g} : A \to \mathbb{R}$, $\dfrac{f}{g}(x) = \dfrac{f(x)}{g(x)}$, for $x \in A$, given than $g(x) \neq 0$ for all $x \in A$.

§4.9 Exponential

A function f such that $x \overset{f}{\mapsto} e^x$ or more compactly $f(x) = e^x$ is called the exponential functional. A pedantic point much in spirit of §4.4 is that $f(x)$ is not a function, it is a value. We'll

henceforth sometimes omit this, but also refrain ourselves from going as far as to call "$y = f(x)$" a function.

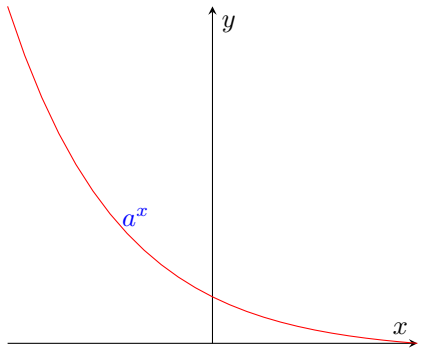
Whenever we write the function a^x we assume two implicit conditions, 1) that $a > 0$, a is non-negative, 2) and that $a \neq 1$.

The reason for adopting such convention is that a^x has a nice domain. If a were to be negative, defining the domain would be quite annoying. It also helps in ensuring that a^x is not a constant function, the cases $a = 0$ and $a = 1$.

Thus, for the function a^x, $a \in (0, 1) \cup (1, \infty)$. This divides the values of a into two intervals. In-fact for these intervals, a^x behaves differently!

§4.9.1 $0 < a < 1$

The function a^x is in such case decreasing. A function is decreasing if $f(x_1) \geqslant f(x_2) \iff x_1 \leqslant x_2$. It is possible to draw the graph of the function by considering values of x for some specific a. But I'll just show the graph. Also, it is decreasing because if $0 < a < 1$, then it is easy to see that for larger values of x, a^x decreases.

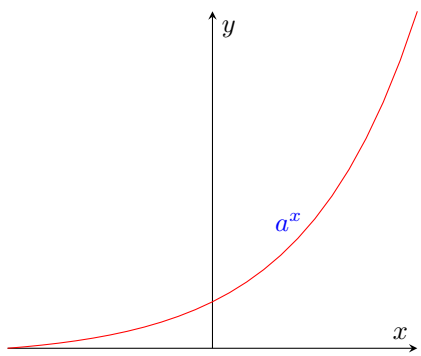


Clearly, for $0 < a < 1$, $a^{g(x)} \geqslant a^{f(x)} \iff g(x) \leqslant f(x)$. The inequality flips. in some manner. This is simply because of the earlier stated decreasing nature of the function.

Inequalities of such kind are called exponential inequalities. The domain $D_f = \mathbb{R}$ and the range is $R_f = \mathbb{R}^+ = (0, \infty)$.

§4.9.2 $a > 1$

Contrastingly to our above function, this function is increasing, for $x_1 \geqslant x_2$ it is easy to see that $a^{x_1} \geqslant a^{x_2}$.



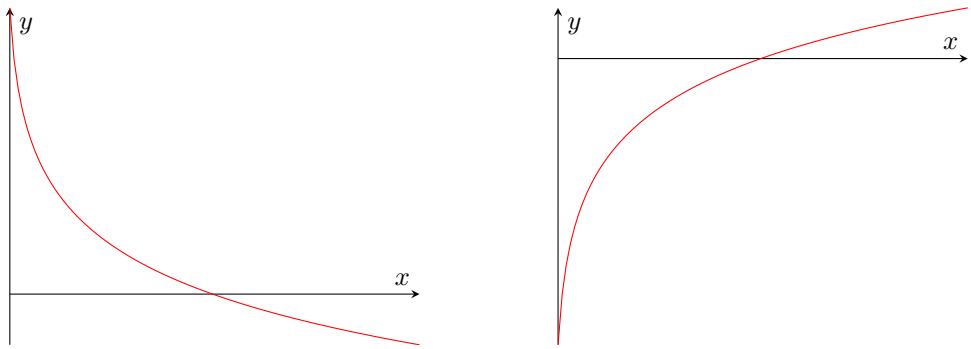
Also, clearly here, $a^{g(x)} \geqslant a^{f(x)} \iff g(x) \geqslant f(x)$. Simply because the function is increasing. The inequality remains the same when comparing exponents. The domain and range are the same as above.

Remark 4.9.1. Note that the usual properties of exponents that we are familiar with, $(a^b)^c = a^{bc}$ and of the same kind only work for b, c being positive integers. For example, consider $(-8)^{1/3} = (-8)^{2/6} = 64^{1/6} = 2$ but clearly $-8^{1/3}$ is -2. Beware.

§4.10 The Logarithm function

The function f of the form $f(x) = \log_a x$ is a logarithmic function. Much like the exponential function, it is defined for $a > 0$ and $a \neq 1$. Unlike the exponential, however, it also has the condition $x > 0$. In-fact, $\log_a x = y \iff a^y = x$. Clearly, from here we have $x > 0$. We may again divide up it into two intervals.

Briefly, for $0 < a < 1$, it is a decreasing function. You can check that for some discrete values. For $a > 1$ it is an increasing function. The domain in both cases is $D_f = (0, \infty)$ and the range is $R_f = \mathbb{R}$.



(a) $0 < a < 1$ (b) $a > 1$

Anyways, the same thing happens with inequalities as it did with the exponential, for $\log_a f(x) \geqslant \log_a g(x)$, if $0 < a < 1$, $f(x) \leqslant g(x)$. And if $a > 1$, $f(x) \geqslant g(x)$.

Also,

$$\begin{cases} \log_a x \geqslant 0 \iff x \geqslant 1 \mid a > 1 \\ \log_a x \geqslant 0 \iff 0 < x \leqslant 1 \mid 0 < a < 1 \end{cases}$$

Proposition 4.10.1 — Some properties of logarithm for $a \in \mathbb{R}_{>0} \setminus \{1\}$. and $m, n \in \mathbb{R}$.

1. $\log_c(ab) = \log_c |a| + \log_c |b|$
2. $\log_c \dfrac{a}{b} = \log_c |a| - \log_c |b|$
3. $\log_c a^b = b \log_c a$
4. $\log_{c^b} a = \dfrac{1}{b} \log_c a$
5. $\log_c p = \dfrac{1}{\log pc}$
6. $a^{\log_c b} = b^{\log_c a}$

§4.11 Some other functions

§4.11.1 Greatest Integer Function

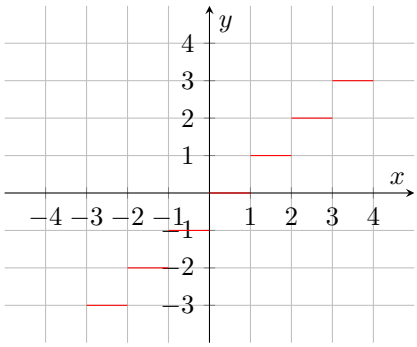
The greatest integer function or the floor function, $\lfloor \bullet \rfloor : \mathbb{R} \to \mathbb{Z}$ is defined as,

Definition 4.11.1 (Floor)

$$\lfloor x \rfloor = a \mid a \leqslant x < a+1, a \in \mathbb{Z}$$

The symbol for floor/GIF is for some reason, mostly [•] in JEE. In any case, they generally mention the symbol in the question.

Therefore, for $x \in [a, a+1)$, where $a \in \mathbb{Z}$, $\lfloor x \rfloor = a$. Using this definition, and setting up values of a, we have the following graph,



The end points of each *step* are open.

Some properties of floor are,

Proposition 4.11.2 — For $x \in \mathbb{R}$ and $n, a, b \in \mathbb{Z}$,

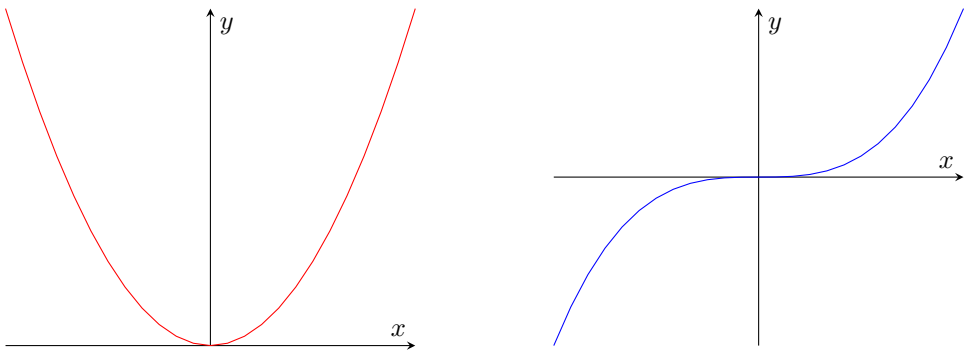
1. $\lfloor x \rfloor + \lfloor -x \rfloor = 0 \mid x \in \mathbb{Z}$
2. $\lfloor x \rfloor + \lfloor -x \rfloor = -1 \mid x \notin \mathbb{Z}$
3. $a \leqslant \lfloor x \rfloor \leqslant b \iff a \leqslant x < b+1$
4. $\lfloor x \rfloor > n \iff x \geqslant n+1$

§4.12 Injective, Surjective and all that

§4.12.1 Injective

If for all $x \in A$, for function $f : A \to B$, x has a 'distinct' image, the function is called one-one or injective. What it means is that any image has only one pre-image under f. Therefore, $f(a) = f(b) \iff a = b$. Since, $f(a)$ and $f(b)$ must have only one pre-image. Since they're the same, the pre-image must also be the same.

If this is not the case, any image may have more than one pre-image under f, the function is not injective and is called many-one. Here, $f(a) = f(b) \not\Longleftrightarrow a = b$.



(a) x^2, a many-one function (b) x^3, a one-one function

Figure 4.8: One-one and Many-one functions

To determine whether a function is one-one or many-one, we may either draw the graph of the function, and check if it is one-one by drawing a horizontal line, if it cuts it at more than one place, then the function is clearly many-one.

The other method is to check whether $f(x_1) = f(x_2) \iff x_1 = x_2$ but is usually difficult. The last method uses differentiation and should be skipped for now, for a continuous function, $\forall x$, $f'(x)$ must have the same sign, i.e, it must be either $\geqslant 0$ or $\leqslant 0$. The solutions of $f'(x) = 0$ must not form an interval. Do not use for functions not continuous over $\mathbb{R}$.

> **Remark 4.12.1.** If a function $f : A \to B$ is injective, then $n(A) \leqslant n(B)$. Thus, B is *at least* as large as A.

§4.12.2 Surjective

Let f be a function, $f : A \to B$. If the range of f is equal to B, the co-domain, then f is surjective. The basic idea is that,

$$\forall y \in B, \exists x \in A \mid f(x) = y$$

So, each $y \in B$ has a pre-image. Functions that are surjective are also called *onto*. The functions that are not surjective are *into*.

To determine if a function is surjective, we may use two methods. One is to figure out the range, if it is the same as the co-domain then it is surjective. The other method is to show that $f(g(y) = y)$. If now we show that $g(y)$ is defined for all $y \in B$ and $g(y) \in A$. Then, f is surjective.

> **Remark 4.12.2.** If $f : A \to B$ is surjective, then $n(A) \geqslant n(B)$. The size of B is at most equal to A.

§4.12.3 Bijection

If a function is both injective and surjective, then it is bijective. Formally, $\forall y \in B$ there is *exactly* one $x \in A$ such that $f(x) = y$.

From the earlier remarks, you may have noted an interesting thing. f being a bijection implies that $n(A) \geqslant n(B)$ and $n(B) \geqslant n(A) \iff n(A) = n(B)$. Thus, two sets are equal if and only if there exists a bijection between them.

§4.13 Infinite sets and their sizes

Extra

Two sets, A and B are equal if and only if there exists some $f : A \to B$ such that f is a bijective. This allows us to extend the definition of equality to infinite sets.

Consider the sets $\mathbb{Z}$ and $\mathbb{N}$. Let $f : \mathbb{N} \to \mathbb{Z}$ such that $f(x) = x$ if x is odd and $-x$ if x is even. Such a f is a bijection!

Thus the cardinalities of these two sets are equals. There are as many integers as there are naturals. And infact, there are as many rationals as there are naturals as well. These sets are *countably* infinite. Contrastingly, the set of reals is larger than the set of naturals and it is uncountably infinite. $n(\mathbb{N}) = n(\mathbb{Z}) = n(\mathbb{Q}) < n(\mathbb{R})$.

In general, if for an infinite set A, $n(A) = n(\mathbb{N})$, A is countably infinite and uncountably infinite if $n(A) \geqslant n(\mathbb{R})$.

§4.14 Even-Odd

§4.14.1 Even Functions

If $\forall x \in D_f, f(-x) = f(x)$ then $f(x)$ is said to be an even function. Clearly, all such functions are many-one. Also, $(x, y) \in f \iff (-x, y) \in f$.

Example 4.14.1 (Even fucntions)

$|x|$, x^2, $\cos(x)$ are even functions.

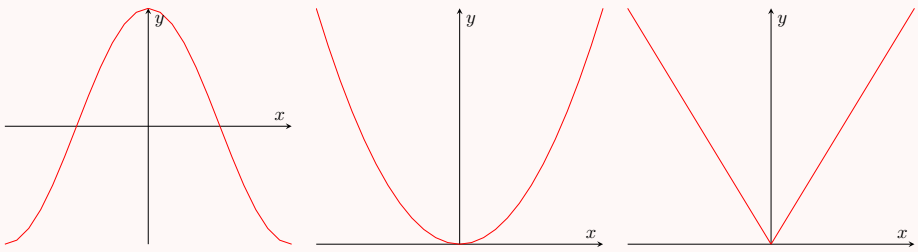


Figure 4.9: Even Functions

§4.14.2 Odd Functions

If $\forall x \in D_f$, $f(-x) = -f(x)$, then f is said to be an odd functions. These functions may or may not be injective. Also, $(x, y) \in f \iff (-x, -y) \in f$.

Example 4.14.2

$\sin(x)$, x^3, $\tan(x)$ are all odd functions.

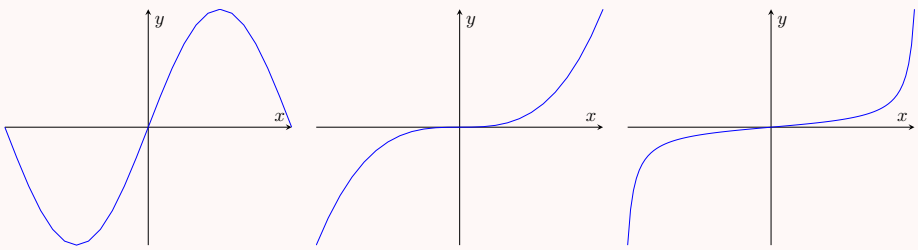


Figure 4.10: Odd Functions

Proposition 4.14.3 — Some properties related to odd and even functions are,

- A function can be even or odd or both or neither.

- f such that $f(x) = 0$ is both even and odd.
- It is possible to express any function as sum of even and odd functions.

$$f(x) = \underbrace{\left(\frac{f(x) + f(-x)}{2}\right)}_{even} + \underbrace{\left(\frac{f(x) - f(x)}{2}\right)}_{odd}$$

- The first derivative of an even function is an odd function and vice versa.
- If an odd function f is continous at 0, then $f(0) = 0$.
- $f'(x) = 0$ at $x = 0$ if f is continous and differentiable at 0.

§4.14.3 Odd and even extensions of a function

If a function f is defined only on $(0, L)$, then we may extend f as an even function as,

$$f(x) = \begin{cases} f(x) & x \in (0, L) \\ f(-x) & x \in (-L, 0) \end{cases}$$

We may extend it as odd function as,

$$f(x) = \begin{cases} f(x) & x \in (0, L) \\ -f(-x) & x \in (-L, 0) \end{cases}$$

§4.15 Periodic Functions

A function f is periodic if $\forall x \in D_f, f(x) = f(x + T)$. The least positive value of T for which this is valid is called the *principal* period of f.

Clearly, $f(x) = f(x + T) = f(x + 2T) = \cdots = f(x + nT)$. One thing of note is that some functions may be periodic but have no fixed period. In particular the constant function, $f(x) = c$ is periodic but there is no findable smallest value for which it is periodic.

§4.15.1 Period of transformed functions

We make the following claims,

Claim 4.15.1 — For a function f of period T, the transformed functions $af(x)$, $f(x \pm a)$, $f(x) \pm a$ all have period T.

Proof. By definition, $f(x) = f(x + T)$. Thus,

$$\begin{aligned} & f(x) = f(x + T) \\ \iff & af(x) = af(x + T) \end{aligned}$$

The rest can be shown similarly. □

Claim 4.15.2 — For a function f of period T, the function $f(ax)$ has period $T/|a|$.

Proof. Obviously, $f(x) = f(x + T)$. Replace x by ax. Then,

$$\begin{aligned} f(ax) &= f(ax + T) \\ f(ax) &= f(a(x + T/a)) \end{aligned}$$

Now, clearly T/a is a period of $f(ax)$. However, the principal period of $f(ax)$ must be $T/|a|$ since a may be negative. □

§4.15.2 Period of Algebraically operated functions

If period of f is $p = m/n$, and period of g is $q = r/s$, then the period of any algebraic manipulation of the two functions, $f+g$, $f-g$, $f \cdot g$ and f/g is $t = \operatorname{lcm}(p, q)$,

$$t = \frac{\operatorname{lcm}(m, r)}{\gcd(n, s)}$$

Where $n, m, r, s \in \mathbb{N}$ and $\gcd(m, n) = \gcd(r, s) = 1$. If p, q are irrational, then their lcm is only defined if they have a common surd. In such a case, $p = a\sqrt{n}$, $q = b\sqrt{n}$ where n is not a perfect square and $a, b \in \mathbb{Z}$. Then, $\operatorname{lcm}(p, q) = \sqrt{n}\operatorname{lcm}(a, b)$.

§4.16 Composite Functions

Let A, B, C be three non-empty sets. Then for $f : A \to B$ and $g : B \to C$, there exists $f \circ g : A \to C$ where $g \circ f(x) = g(f(x))$, $\forall x \in A$.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend right, "g \circ f"'] & B \arrow[r, "g"] & C \end{tikzcd}$$

Figure 4.11: Composite Functions

Thus, the image of $x \mid x \in A$ under $g \circ f$ is the image under g of the image of x under f. Clearly, $R_f \subseteq D_{g \circ f}$. Note that in general $g \circ f \neq f \circ g$.

Also, composition is associative, $g \circ (f \circ h) = (g \circ f) \circ h$.

Example 4.16.1

If $f : \mathbb{R} \to \mathbb{R} \mid f(x) = \sin(x)$ and $g : \mathbb{R} \to \mathbb{R} \mid g(x) = \cos(x)$.

Then $g \circ f = \cos(\sin(x))$ and $f \circ g = \sin(\cos(x))$.

§4.16.1 Injectivity and Surjectivity of Composite functions

From the careful observation of composite functions, we may propose,

Proposition 4.16.2 — If $f : A \to B$, $g : B \to C$ are injective, then $g \circ f : A \to C$ is also injective.

Proof. Note that $f(a) = f(b) \iff a = b$ and $g(a) = g(b) \iff a = b$, then,

$$\begin{aligned} g \circ f(a) = g \circ f(b) &\iff g(f(a) = g(f(b))) \\ &\iff f(a) = f(b) \\ &\iff a = b \end{aligned}$$

Thus, $g \circ f(a) = g \circ f(b) \iff a = b$. □

Proposition 4.16.3 — If $f : A \to B$ and $g : B \to C$ are surjective, then $g \circ f$ is also surjective.

Proof. Given an arbitrary element $c \in C$, $\exists x \in B, g(x) = c$. Now, $\forall x \in B, \exists a \in A, f(a) = x$. Substitute $x = f(a)$ and thus for $\forall b \in C, \exists a \in A \mid g \circ f(a) = c$. □

PART C

Appendices

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A Logic and Proofs

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